Math 344, S19 **Name:** Answer

Quiz 1

<u>Instructions:</u> There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (7 points) Given $\mathbf{v} = (3, 4)$ and $\mathbf{w} = (12, -5)$.

(a) Find the *cosine* of the angle θ between $\mathbf{v}$ and $\mathbf{w}$.

$$\|v\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

$$\|w\| = \sqrt{12^2 + (-5)^2} = \sqrt{169} = 13$$

$$v \cdot w = (3, 4) \cdot (12, -5) = 3 \cdot 12 + 4 \cdot (-5) = 16$$

$$\cos\theta = \frac{v \cdot w}{\|v\|\,\|w\|} = \frac{16}{5 \cdot 13} = \frac{16}{65}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{16}{65}\right)$$

(b) Find a number c so that $\mathbf{w} - c\mathbf{v}$ is perpendicular to $\mathbf{v}$. (*Hint:* using $(\mathbf{w} - c\mathbf{v}) \cdot \mathbf{v} = 0$)

$$(w - cv) \cdot v = 0$$

$$w \cdot v - c\, v \cdot v = 0$$

$$c\,\|v\|^2 = w \cdot v$$

$$c = \frac{w \cdot v}{\|v\|^2} = \frac{16}{5^2} = \frac{16}{25}$$

Problem 2. (3 points) Compute $\mathbf{Ax}$ (i.e., the matrix $\mathbf{A}$ multiplies the vector $\mathbf{x}$) with

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & 0 \\ -3 & 1 & -1 \\ -4 & 1 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

by using linear combination of the columns of $\mathbf{A}$ with components of $\mathbf{x}$ as coefficients.

$$Ax = \begin{bmatrix} 1 & -2 & 0 \\ -3 & 1 & -1 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

$$= 2\begin{bmatrix} 1 \\ -3 \\ -4 \end{bmatrix} + (-1)\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} + 3\begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ -6 \\ -8 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 2+2+0 \\ -6-1-3 \\ -8-1+6 \end{bmatrix} = \begin{bmatrix} 4 \\ -10 \\ -3 \end{bmatrix}$$